

Findings Summary: Does the Committed-Minority Framework Apply to LLM Multi-Agent Systems?

Framework under test: Xie, Sreenivasan, Korniss, Zhang, Lim & Szymanski (2011), *Social consensus through the influence of committed minorities* (PRE 84, 011130) — the committed-minority tipping point $p_c \approx 10\%$ in the binary-agreement model.

Thesis (one line): *Heterogeneity helps conditionally, not automatically. Its benefit is bounded by error correlation (LLMs are far more correlated than their lineage diversity suggests), gated by a competence floor, and realized only by aggregators that can exploit decorrelation — but on open-ended deliberation it wins cleanly by breaking groupthink.*

The professor's three questions, answered

RQ1 — Homogeneity vs Heterogeneity: how much more vulnerable are monocultures?

Answer: more vulnerable to common-mode failure, but the heterogeneity protection is quantitatively bounded — and can backfire. - **Groupthink (measured):** in value deliberation a homogeneous panel (6× claude-sonnet) collapsed to *identical* stances (final stance spread **0.00**) with near-verbatim reasoning; a 5-architecture heterogeneous panel kept genuine spread (**0.47**), raised more distinct considerations (**6.9 vs 5.9**), and resisted a committed extremist better (**sway 0.13 vs 0.20**). - **Bounded protection (measured):** across 7 independent lineages, error correlation $\rho \approx 0.53$, effective independent voters $N_{\text{eff}} \approx 1.7$ of 7. Different-lineage models still fail on the *same* questions → diversity buys far less than the count implies. Diversity headroom only **+9%** (90% ≥1-right vs 81% best single); **10%** of items defeat all seven. - **Confirmed at the FRONTIER** (100 hard Qs incl. GPT-5, Claude-Opus): mean $\rho = 0.48$, $N_{\text{eff}} = 1.8$ of 7 — the illusion of cognitive diversity holds for the very best models too. And as the best single model strengthens (claude-opus = 92%), the diversity headroom **collapses to +2%** (95% ≥1-right vs 92% best single): a dominant model already covers what diversity would add, so heterogeneity offers even *less* on factual tasks at the top. - **Backfire (measured):** an unmatched diverse pool *lost* to a strong monoculture (72% vs 75%) via competence dilution (9 dilution-losses vs 5 decorrelation-wins). - **Even matched + decorrelated + smart aggregation didn't net-win on factual:** homo claude-sonnet×3 = 86% vs hetero[sonnet+qwen235b+llama70b] = 80% majority / 82% with a verify-style judge (sonnet solo 81%). The judge recovered some coverage (80→82) as theory predicts, but sonnet so outclasses its peers that dilution wins. **On factual correctness, a strong monoculture is hard to beat with diversity.**

RQ2 — Model scale: how does size affect robustness/susceptibility?

Answer: scale acts as a capability gate / competence floor, not a smooth curve. - **Capability gate:** below ~30B, models cannot execute the basic primitives — naming-game rule-following (3-4B ≈ 33-50%) and GovSim cooperation (weak models collapse the commons even with zero defectors). They are systemically broken substrates regardless of composition. - **Competence floor ($a > 0.5$):** below chance, a model degrades any majority ensemble — homogeneous *or* heterogeneous. Above it, scale → accuracy → useful contribution. - **Reasoning ≠ better here:** reasoning models (gpt-5-nano 33%) underperform fast non-reasoning instruct models on these coordination tasks.

RQ3 — Tipping points: is there a quantifiable threshold?

Answer: yes — two distinct thresholds. - **Committed-minority tipping point ≈ 10%** (the prof's framework) reproduced for capable LLMs across 5 lineages (DeepSeek 0.084 → gpt-4o-mini 0.118; analytic 0.0979), with intrinsic option-bias controlled. Below the capability gate it is undefined (small models can't sustain the dynamics). - **Condorcet floor ($a = 0.5$)** for ensemble benefit (our theory): the threshold at which adding agents flips from helpful to harmful. - **Cooperation-collapse tipping** (GovSim, LLM): the committed-minority tipping point GENERALIZES from opinion to cooperation, at a DIFFERENT critical fraction. A frontier society (claude-sonnet) sustains the commons up to **20% defectors** (10/10 months, 0% collapse) and **collapses at 30%** (100%) — a cooperation tipping point at **~25%**, close to Centola's 25% social-convention threshold and distinct from the ~10% opinion threshold. Below the cooperation capability floor (gpt-4o-mini) the commons collapses even at $p=0$.

The theoretical contribution: the competence band (THEORY_competence_band.md)

A model joins an ensemble as asset or liability based on **accuracy a** and **error correlation ρ** : 1. **Floor:** $a < 0.5$ → bad model everywhere (Condorcet). 2. **Two equal models > one** by $N_{\text{eff}} = 2/(1+\rho)$; LLM $\rho \approx 0.53$ → ≈ 1.3 effective. 3. **Two equally-incompetent correlated models $\nrightarrow$ one** (and can be worse). 4. **The aggregator decides realization:** diversity creates a *coverage ceiling* ($\propto 1-\rho$); majority voting captures it only above the floor, while verify/select/ deliberate aggregators capture it whenever a correct answer is merely *present*. - *Empirical:* three sub-floor weak models ($a \approx 33-38\%$, $\rho=0.29$) → majority 36% (= a single model's panel), but coverage 60% vs 38% single. So "two weak > one weak" is TRUE for verify/select use cases, FALSE for majority voting.

Paradigms (multi-domain validation)

Paradigm	Domain	Heterogeneity effect	Statu:
Naming game	opinion contagion	$p_c \approx 10\%$ transfers; capability-gated; blind to heterogeneity by construction	done
Adversarial debate	factual problem-solving moral	bounded by ρ ; net-win needs matched+decorrelated+good aggregator	done; matched+j running

Value deliberation	dilemmas (no ground truth)	heterogeneity wins: more perspectives, less groupthink, resists extremist	done
GovSim	commons cooperation	frontier sustains, defector- minority collapses it; capability-gated	done (LLM)

Honest open items (for a polished submission)

- Wider question sets / more seeds + CIs to tighten ρ , p_c , and the GovSim tipping fraction.
- Pin the GovSim cooperation tipping point (sweep $p=0.1/0.2/0.3$ with a frontier model).
- Explicit prompt-injection propagation (homo vs hetero) for the adversarial-susceptibility angle.
- Statistical tests on the deliberation diversity/sway deltas.

Core empirical program complete: all four paradigms run, theory stated and tested.

Artifacts

RESULTS_error_correlation.md · THEORY_competence_band.md ·
 DELIBERATION_DYNAMICS.md · CASE_STUDIES.md · ARCHITECTURE.md ·
 DESIGN.md